Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/06/2024
PI Name:	Lewandowski, Beth Ph.D.		
Project Title:	Tempus ALS ISS Technology Demonstration		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	beth.e.lewandowski@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	216-433-8873
Organization Name:	NASA Glenn Research Center		
PI Address 1:	Bioscience and Technology Branch		
PI Address 2:			
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	08/08/2022	**End Date:**	09/30/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Lemery, Jay	**Contact Phone:**	
Contact Email:	jay.lemery@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Adding Co-I names who have been working on the project: Courtney Schkurko Kimesha Calaway Rachael Miller		
COI Name (Institution):	Lehnhardt, Kris M.D. (NASA Johnson Space Center) Easter, Benjamin M.D. (NASA Johnson Space Center) Suresh, Rahul M.D. (NASA Johnson Space Center) Thompson, Moriah M.D. (NASA Johnson Space Center) Schkurko, Courtney (NASA Glenn Research Center) Calaway, Kimesha (ZIN Technologies, Inc.) Miller, Rachael (KBR, Inc.)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The aim of this project is to evaluate the use of the Tempus Advanced Life Support (ALS) device in the spaceflight environment: • In collaboration with the European Space Agency (ESA), Phase 1 entails using the Tempus ALS device, compared to current International Space Station (ISS) medical devices, for the completion of Periodic Health Status exams and ISS medical contingency drills for 6 crewmembers (ESA = 2, NASA = 4), in order to achieve ESA's test objectives as described in the ESA Tempus ALS Technology Demonstration Activity Requirements Document (ARD)1. • NASA-led Phase 2 of the demonstration will entail Exploration Medical Capability (ExMC) and the Exploration Medical Integrated Product Team (XMIPT) working together to demonstrate the use of Tempus ALS on the ISS as part of progressively Earth-Independent Medical Operations (EIMO) through the execution of onboard medical simulations that emphasize crew autonomy during exploration medical care. The Tempus ALS is a multifunctional medical device with capabilities that include vital sign collection and monitoring, communications to enable telemedicine, interfaces with electronic health records, ultrasound, video laryngoscopy, and defibrillation. Tempus for ISS project activities will include evaluation of the usability of the Tempus ALS in a spaceflight environment, its functionality during the diagnosis and treatment of simulated medical conditions, and the increased onboard capabilities that it can provide when operated autonomously. The project results will be used to answer the question: Is the Tempus ALS a good candidate for inclusion within spaceflight medical system operations on exploration missions? For this project, simulations of one or more medical contingency scenarios will be executed and ISS astronauts will be required to follow and perform the NASA and ESA developed medical procedures necessary for diagnosing and treating the medical emergency. The astronauts will be required to use several of the capabilities of the Tempus ALS device while performing these procedures. The astronauts will perform the procedures under varying levels of autonomy, including with and without communication delays and with different types of procedural guidance.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. Currently aboard the International Space Station (ISS), NASA has the ability to perform in-flight medical vital signs monitoring, in-flight imaging, and Periodic Health Status exams with a suite of medical devices and techniques. On the ISS, these medical procedures are currently accomplished with reliance on the Flight and Medical Operations teams on the ground. For long-duration space missions, mass, power, and volume requirements for medical devices will be increasingly limited. Further, with increased distance, the Earth-Independent Medical Operations (EIMO) capability will be a cornerstone of the provision of adequate medical care during exploration missions beyond low-Earth orbit (BLEO). There is a need to identify and evaluate medical devices that will be amenable to the EIMO construct.
Research Impact/Earth Benefits:	The Tempus ALS device is designed for operations in pre-hospital and in remote settings. The Tempus ALS consists of two devices, the Tempus Pro (for vital sign measurements and clinical imaging), and the Tempus LS (for atrial defibrillation). The demonstrations of the Tempus Pro device performed during this study will provide additional user experience information as the device is used in the remote and extreme environments of the ISS and the Exploration Atmosphere Chamber. This information could further aid its effective use in other remote and extreme Earth-based environments.
Task Progress:	NOTE: This project continues as "Tempus ALS ISS Technology Demonstration" under a new Principal Investigator (PI): Jerry Myers, Ph.D. The previous PI, Dr. Beth Lewandowski, has retired from NASA (Ed., 6/3/25). For current reporting, see the new project under Dr. Myers. During the fiscal year 2024, the Tempus Pro was demonstrated along with another Multi-functional Integrated Medical (MIM) device, the LifeBot, by non-medically trained participants within the ground-based ExMC Technology Demonstration Test Bed, located at the NASA Glenn Research Center. The participants followed text-based procedures that led them through demonstrations of the MIM device's functionality, including vital sign measurement, the use of several different types of scopes and ultrasound imaging with different types of ultrasound probes. The participants demonstrated the vital sign functionality by taking measurements of themselves and demonstrated the scope and ultrasound functionality on another participant acting as a patient. The participant's workload was assessed by collecting their heart rate during the performance of the demonstration activities, collection of a NASA Task Load Index (TLX) survey after each activity and by recording the amount of time necessary to perform each activity. The participants also filled out user experience surveys, which collected information about the ease of use of the devices. The device functionality was also compared to the exploration medical system requirements and the clinical utility of the ultrasound images resulting from the devices is also being assessed. An internal NASA report has been produced which provides more details and the key findings from the ground demonstrations.
Bibliography Type:	Description: (Last Updated: 06/06/2024)
Abstracts for Journals and Proceedings	Miller RS, Lewandowski BE, Schkurko CM, Calaway KM, Hew YM, Garbino A, Stratton ER, Thompson MS, Suresh R, Yang JD, Valentine RW. "Exploration atmosphere demonstration of a multi-functional integrated medical device (Tempus Pro)." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Lewandowski BE, Schkurko CM, Calaway KM, Miller RS, Easter BD, Suresh R, Lehnhardt KR, Thompson MS. "A multi-faceted approach to demonstrating multi-functional integrated medical devices to advance Earth-independent medical operations." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstract. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024